Richard Franks

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PROFESSIONAL SUMMARY

Principal / Distinguished-level full-stack engineer with 15 years of experience building production web applications, APIs, and domain-specific software across healthcare, fintech, and scientific-adjacent environments. Strong hands-on background in Python, Java, TypeScript, React, RESTful services, SQL/NoSQL-backed systems, and AI-enabled workflows. Especially effective in roles that require deep technical ownership, rapid ramp-up in complex domains, and close collaboration with cross-functional experts to turn difficult workflows into reliable, user-centered software.

CORE SKILLS

Languages: Python, Java, JavaScript, TypeScript
Frontend & Full-Stack Frameworks: React, Next.js, NestJS, Spring, Spring Boot, Django, Flask, Hibernate
Backend & Integration: REST APIs, web services, backend services, service integration, full-stack development, monorepo development
Databases & Data Modeling: SQL, MySQL, PostgreSQL, Oracle, CosmosDB, MongoDB, relational and non-relational data modeling
AI & Applied ML: LLMs, RAG, LangChain, AI workflow orchestration, model evaluation, transcription pipelines (STT and TTS)
Cloud & Platform: Azure Functions, AWS, cloud-native application development, third-party API integration
Delivery & Quality: CI/CD, GitHub Actions, Jenkins, automated testing, testing automation, Jest
Leadership & Execution: Architecture, code reviews, technical mentoring, cross-functional collaboration, end-to-end product development

EXPERIENCE

Carebrain — Distinguished Engineer

Remote | Sep 2025 – Mar 2026

- Ramped from a Java/Python background into a full TypeScript / Next.js / NestJS monorepo and shipped substantial product functionality with minimal guidance.
- Built a LangChain-based pipeline with intelligent model selection and answer evaluation, improving answer quality and model choice instead of relying on fixed one-shot prompting to a single vendor preventing vendor lock-in and increasing up-time—if a particular model is down for any reason, the other models that are still available are used. Answers from the model can then be evaluated and appropriate steps taken based on that analysis.
- Implemented an in-person scribe workflow using Deepgram, AssemblyAI, ElevenLabs, AWS Transcribe Medical, and AWS Comprehend Medical, with extraction of topics, keywords, insights and tasks at 95%+ accuracy. Supported speaker diarization and multiple model council.
- Implemented HIPAA audit logging of PHI transmission across 100% of internal APIs, third-party service calls, and database transactions.
- Built a patient list browser allowing a user to browse relevant EHR medical data and view clinical indicators—either pre-computed or computed on-the-fly from static analysis or LLM queries.
- Improved Telehealth transcription workflows to 95%+ accuracy.
- Implemented a slack-like Clinician-to-Clinician Group Chat client built on top of getstream.io.

Commure — Senior Software Architect / Principal Software Engineer

Remote | Jun 2022 – Sep 2025

- Architected and built an AI copilot using RAG and multiple LLMs over clinical EHR data, including summary generation and direct question answering—consisted of a Python service (calling CosmosDB/MongoDB NoSQL stores, creating embeddings and using them to query documents, utilizing LangChain to call multiple LLM models/vendors and performing answer analysis, evaluation and validation.
- Extended the AI copilot system to analyze progress notes, extract complications and risk factors, and support downstream recommendations tied to clinical coding systems—like ICD-10 codes and CPT codes
- Brought applied AI capabilities into production workflows in a highly specialized domain, working across software, data, and end-user needs.

PatientKeeper — Principal Software Engineer / Architect

Waltham, MA | Dec 2013 – Sep 2021

- Rebuilt the portal application framework to support loosely coupled product screens across multiple suites, enabling a more dynamic user experience using an MVC-based architecture using Backbone.js/Handlebars combined with a RESTful Java Spring backend.
- Helped reimagine the EHR with a dashboard framework supporting 200+ robust gadget implementations.
- Created reusable custom components used across the engineering organization—e.g. Patient List and Patient Assignment, which was used across the portal application for patient list, rounding order, and patient selection.
- Led design sessions, code reviews, created diagrams and documentation, mentored engineers, integrated automated tests into CI/CD pipelines.
- Served as an active member of the Architecture Board.
- Led teams of up to 3-5 members.

DressageMarket.com, LLC — Chief Engineer / Entrepreneur

Remote | Aug 2009 – Mar 2021

- Cofounded and served as sole engineer for a web-based classified marketplace, owning product architecture, implementation, and ongoing evolution end to end.
- Worked closely with other cofounders to help work through product roadmap and help materialize vision.
- Built using Python and Django.

REZ-1 — Software Engineering Consultant

Wellesley, MA | Oct 2010 – Jun 2012

- Contributed across full stack on a large-scale enterprise rewrite for freight marketplace software, spanning UI, middle-tier services, backend entities, testing, deployment, and maintenance.
- Worked with Vaadin, Liferay, Tomcat, Spring, Hibernate, JPA, JAX-WS, and jUnit, including contribution to an annotation-driven framework for faster development.

CashStar — Engineer

Portland, ME | Jun 2009 – Oct 2010

- Led engineering on a Java REST service layer connecting a customer-facing application to external gift-card processors.
- Built and certified integrations with major credit card processors and supported launches across 36 national and regional brands.
- Maintained 99.88% uptime while supporting the service in production and strengthening automated testing and CI workflows.

EDUCATION

Boston University, 2003 — B.S., Biomedical Engineering

University of Southern Maine, 2008 — M.S., Computer Science